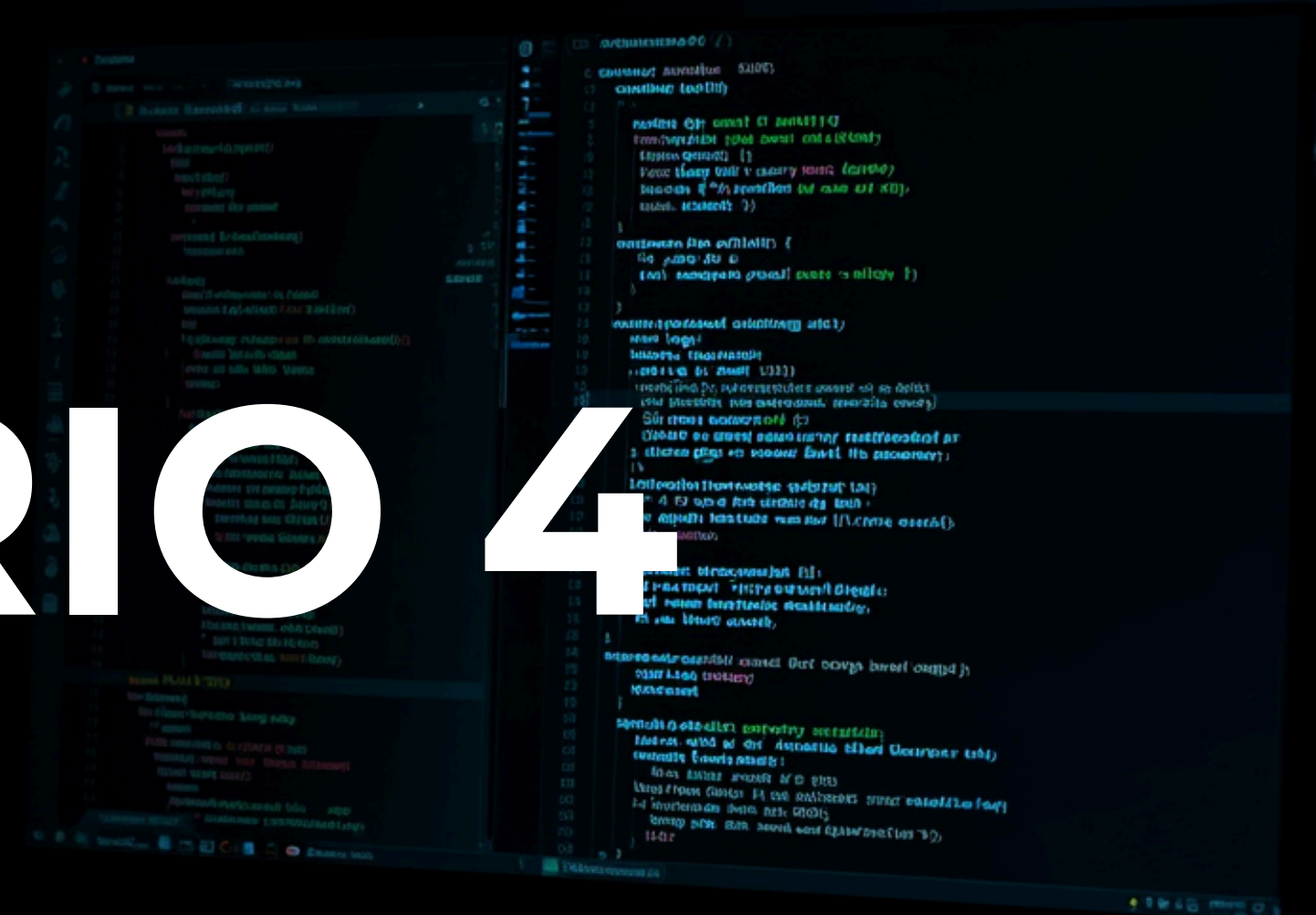
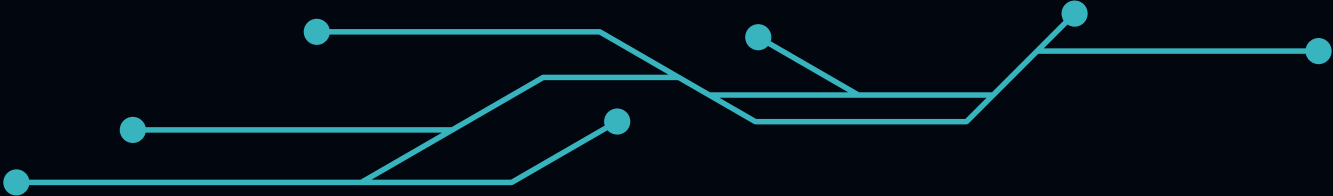




LABORATORIO 4

Hardening y Análisis de Vulnerabilidades con Docker





OBJETIVO DEL LABORATORIO

Implementar un entorno vulnerable utilizando Docker para identificar vulnerabilidades reales y aplicar técnicas de hardening defensivo sobre servicios expuestos.



Análisis de Puertos

Escaneo avanzado y enumeración detallada de servicios activos.



Hardening SSH

Configuración segura del servicio SSH para mitigar ataques.



Automatización y Logs

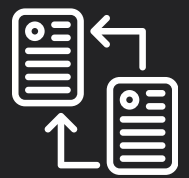
Actualizaciones automáticas y análisis forense de registros.



CREACIÓN DEL ENTORNO

MONTAJE DOCKER

Contenedores: Ubuntu vulnerable (víctima) y Kali Linux (atacante)
Comandos: docker-compose up -d | docker ps



Contenedores



docker-compose



nano ubuntu

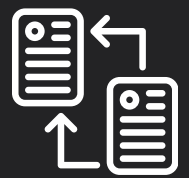
```
GNU nano 8.1 docker-compose.yml *
services:

  ubuntu_vuln:
    build:
      context: .
      dockerfile: Dockerfile.ubuntu
    container_name: ubuntu_vulnerable_lab1
    ports:
      - "2222:22"
      - "8080:80"
      - "2121:21"

  kali:
    image: kalilinux/kali-rolling
    container_name: kali_lab1
    tty: true
    stdin_open: true
```

CONFIGURACIÓN DEL SERVIDOR

Se configuró el contenedor Ubuntu vulnerable instalando servicios inseguros como FTP, SSH y HTTP para posteriormente analizar su exposición.



Contenedores



docker-compose



nano ubuntu

```
kali@kali: ~/lab1_hardening_profundo
GNU nano 8.1 Dockerfile.ubuntu *
FROM ubuntu:20.04

ENV DEBIAN_FRONTEND=noninteractive

RUN apt update && apt install -y \
  openssh-server \
  apache2 \
  vsftpd \
  sudo \
  nano \
  net-tools

RUN mkdir /var/run/sshd

# Configuración vulnerable SSH
RUN echo "PermitRootLogin yes" >> /etc/ssh/sshd_config
RUN echo "root:1234" | chpasswd

# Configurar FTP vulnerable
RUN echo "anonymous_enable=YES" >> /etc/vsftpd.conf

EXPOSE 22 80 21

CMD service ssh start && \
  service apache2 start && \
  service vsftpd start && \
  tail -f /dev/null
```

RECONOCIMIENTO Y ENUMERACIÓN NMAP

Comando ejecutado:

```
nmap --script vuln ubuntu_vulnerable_lab1
```

Identificación de: Puertos abiertos,
Servicios activos, Vulnerabilidades detectadas

El escaneo permitió identificar servicios inseguros
expuestos que posteriormente fueron utilizados en
pruebas ofensivas y hardening.



```
root@4ced45c4302b: /  
File Actions Edit View Help  
root@4ced45c4302b: / x root@4ced45c4302b: / x root@4ced45c4302b: / x  
(kali@kali)-[~/lab1_hardening_profundo]  
$ docker exec -it kali_lab1 bash  
(root@4ced45c4302b)-[/]  
# nmap ubuntu_vulnerable_lab1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-18 15:37 +0000  
Nmap scan report for ubuntu_vulnerable_lab1 (172.24.0.2)  
Host is up (0.000013s latency).  
rDNS record for 172.24.0.2: ubuntu_vulnerable_lab1.lab1_hardening_profundo_defaul  
t  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE  
21/tcp    open  ftp  
22/tcp    open  ssh  
80/tcp    open  http  
MAC Address: 72:2A:FE:1C:6C:20 (Unknown)  
  
Nmap done: 1 IP address (1 host up) scanned in 0.23 seconds  
  
(root@4ced45c4302b)-[/]  
# nmap -sV ubuntu_vulnerable_lab1  
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-18 15:37 +0000  
Nmap scan report for ubuntu_vulnerable_lab1 (172.24.0.2)  
Host is up (0.000015s latency).  
rDNS record for 172.24.0.2: ubuntu_vulnerable_lab1.lab1_hardening_profundo_defaul  
t  
Not shown: 997 closed tcp ports (reset)  
PORT      STATE SERVICE VERSION  
21/tcp    open  ftp      vsftpd 3.0.5  
22/tcp    open  ssh      OpenSSH 8.2p1 Ubuntu 4ubuntu0.13 (Ubuntu Linux; protocol 2.0)  
80/tcp    open  http     Apache httpd 2.4.41 ((Ubuntu))
```


ANÁLISIS DE SERVICIOS DETECTADOS

Identificación de servicios inseguros expuestos y evaluación de riesgos reales.

✓ Análisis detallado por servicio ✓ Evaluación de superficie de ataque



Puerto 21

FTP inseguro
Se comprobó
experimentalmente que FTP
transmite información sin
cifrado.



Puerto 22

SSH activo
Acceso remoto expuesto
a ataques brute force



Puerto 80

HTTP expuesto
Servidor web vulnerable
a ataques remotos

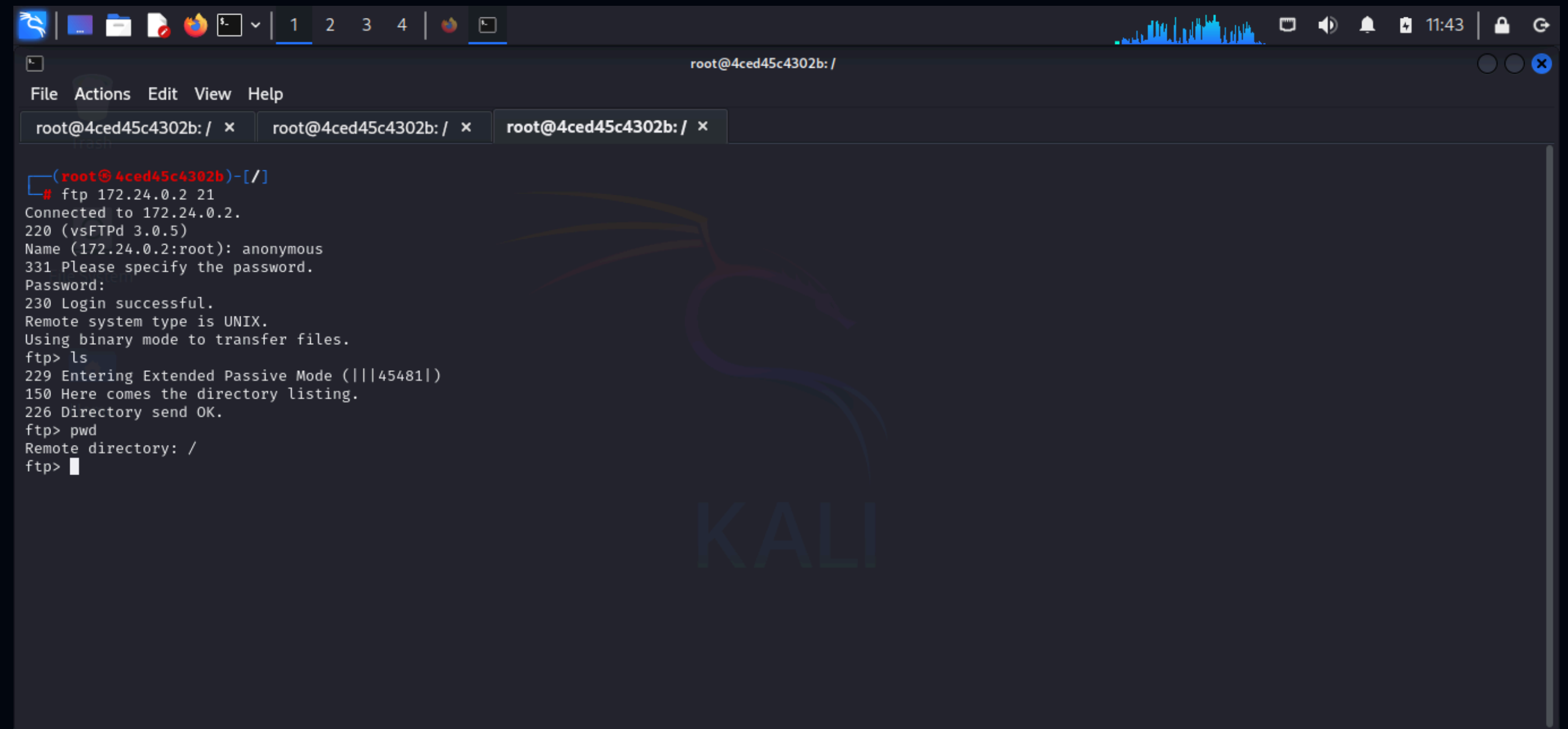


Riesgos

Evaluación detallada de
cada servicio para
priorizar hardening

LOGIN FTP

Se realizó conexión FTP desde Kali Linux hacia el servidor vulnerable, comprobando que el protocolo permite autenticación sin cifrado.



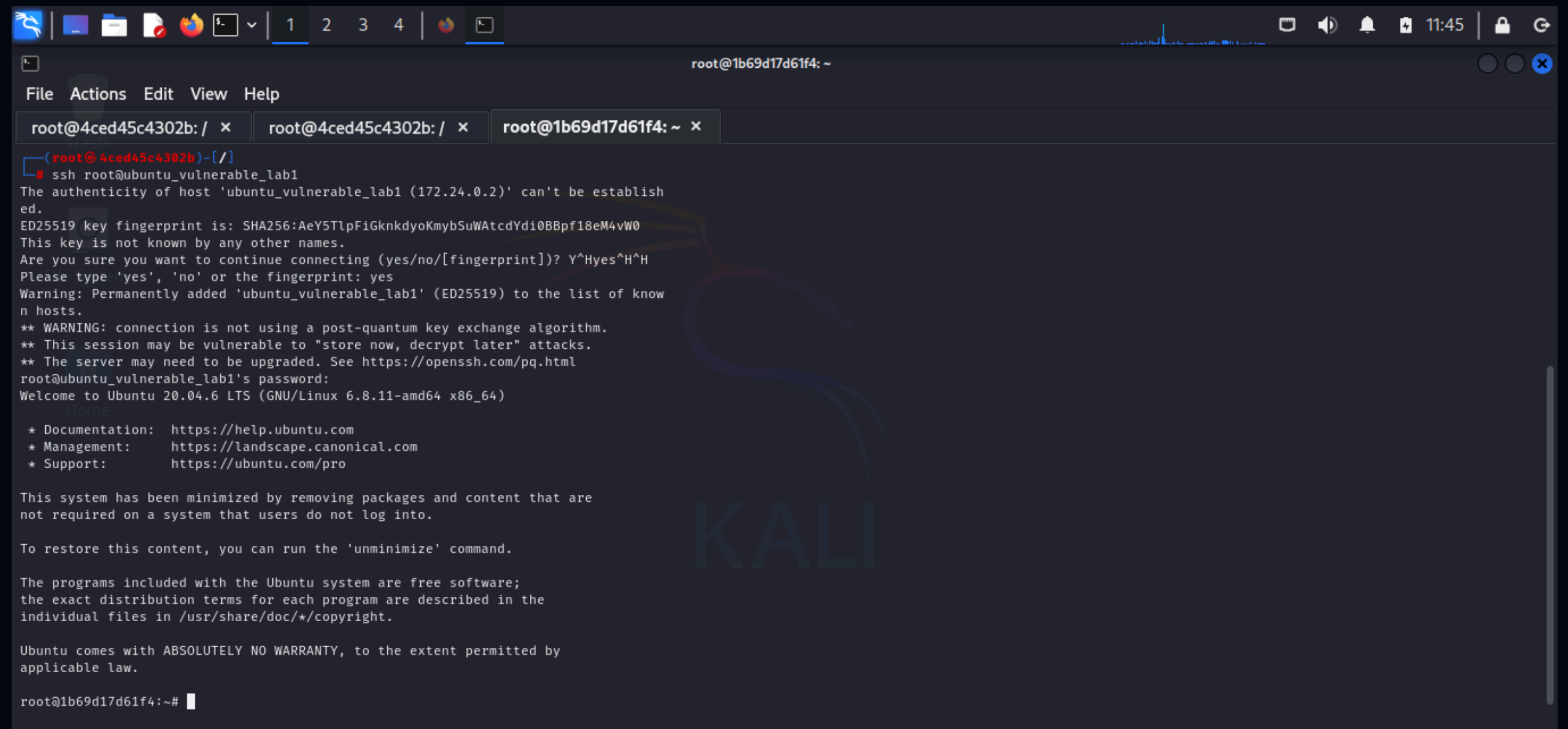
The screenshot shows a Kali Linux terminal window with a dark theme. The window title is 'root@4ced45c4302b: /'. The terminal output shows the following commands and responses:

```
(root@4ced45c4302b)-[/]  
# ftp 172.24.0.2 21  
Connected to 172.24.0.2.  
220 (vsFTPd 3.0.5)  
Name (172.24.0.2:root): anonymous  
331 Please specify the password.  
Password:  
230 Login successful.  
Remote system type is UNIX.  
Using binary mode to transfer files.  
ftp> ls  
229 Entering Extended Passive Mode (||||45481|)  
150 Here comes the directory listing.  
226 Directory send OK.  
ftp> pwd  
Remote directory: /  
ftp> █
```

The terminal window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. There are three tabs open, all titled 'root@4ced45c4302b: /'. The background of the terminal window features a faint 'KALI' logo.

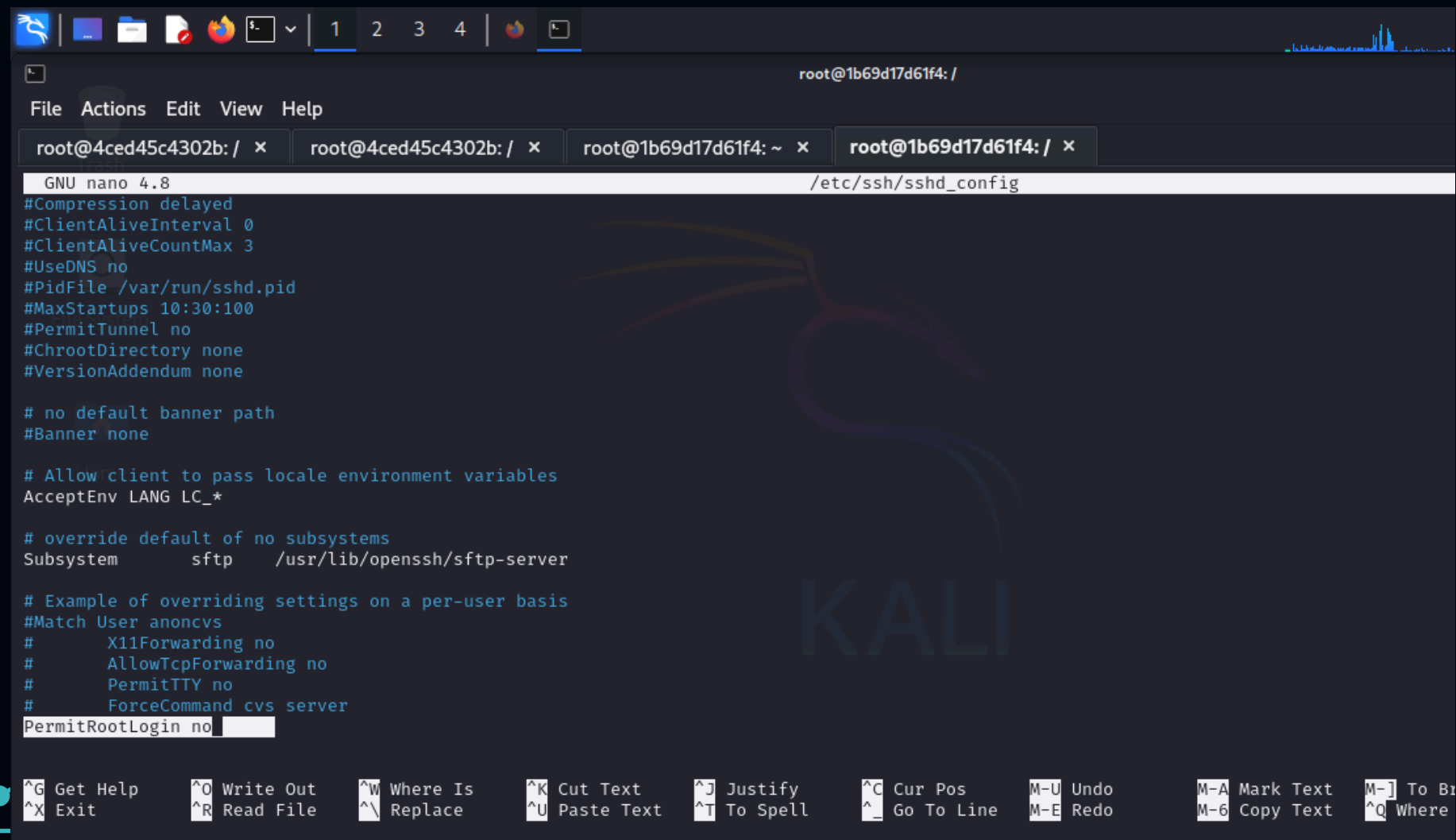
LOGIN ROOT

Se comprobó que el servidor permitía acceso remoto root mediante SSH, representando un alto riesgo de seguridad.



```
root@1b69d17d61f4: ~  
File Actions Edit View Help  
root@4ced45c4302b: / x root@4ced45c4302b: / x root@1b69d17d61f4: ~ x  
(root@4ced45c4302b)-[/]  
# ssh root@ubuntu_vulnerable_lab1  
The authenticity of host 'ubuntu_vulnerable_lab1 (172.24.0.2)' can't be established.  
ED25519 key fingerprint is: SHA256:AeY5TlpFiGknkdyoKmybSuWAtcdYdi0BBpf18eM4vW0  
This key is not known by any other names.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? Y^Hyes^H^H  
Please type 'yes', 'no' or the fingerprint: yes  
Warning: Permanently added 'ubuntu_vulnerable_lab1' (ED25519) to the list of known hosts.  
** WARNING: connection is not using a post-quantum key exchange algorithm.  
** This session may be vulnerable to "store now, decrypt later" attacks.  
** The server may need to be upgraded. See https://openssh.com/pq.html  
root@ubuntu_vulnerable_lab1's password:  
Welcome to Ubuntu 20.04.6 LTS (GNU/Linux 6.8.11-amd64 x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/pro  
  
This system has been minimized by removing packages and content that are  
not required on a system that users do not log into.  
  
To restore this content, you can run the 'unminimize' command.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
root@1b69d17d61f4:~#
```


HARDENING SSH



```
GNU nano 4.8 /etc/ssh/sshd_config
#Compression delayed
#ClientAliveInterval 0
#ClientAliveCountMax 3
#UseDNS no
#PidFile /var/run/sshd.pid
#MaxStartups 10:30:100
#PermitTunnel no
#ChrootDirectory none
#VersionAddendum none

# no default banner path
#Banner none

# Allow client to pass locale environment variables
AcceptEnv LANG LC_*

# override default of no subsystems
Subsystem        sftp        /usr/lib/openssh/sftp-server

# Example of overriding settings on a per-user basis
#Match User anoncvs
#    X11Forwarding no
#    AllowTcpForwarding no
#    PermitTTY no
#    ForceCommand cvs server
PermitRootLogin no
```



PermitRootLogin

Deshabilitar acceso root directo para mayor seguridad.



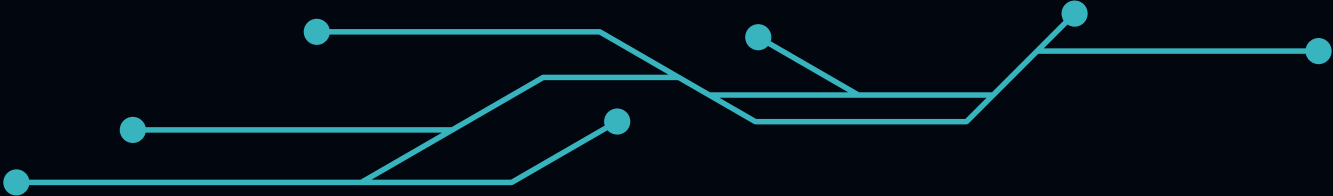
Restricciones

Control de usuarios y métodos de autenticación permitidos.



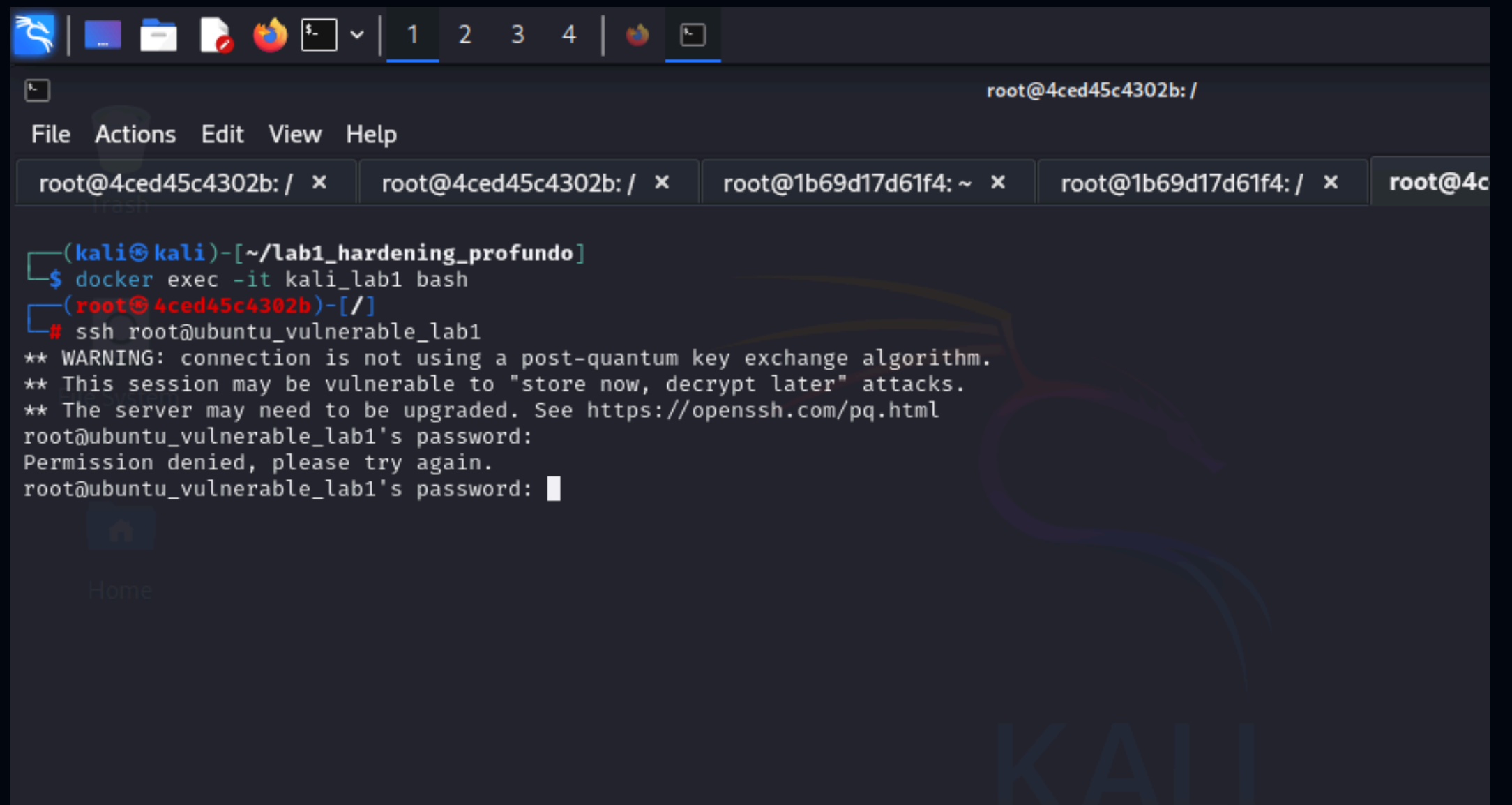
Configuración Segura

Parámetros optimizados para protección del servicio SSH.



LOGIN ROOT DENEGADO

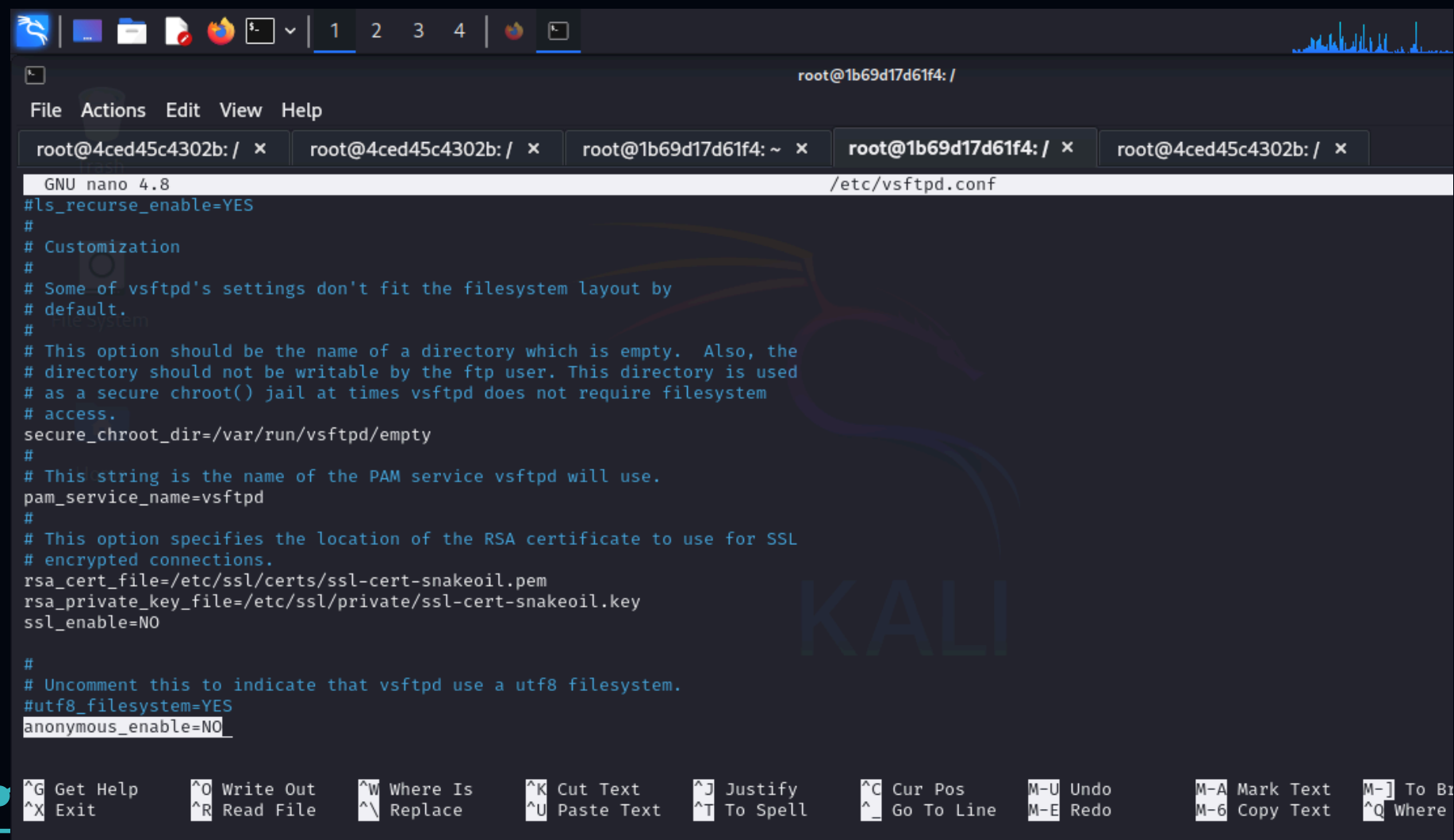
Después de modificar PermitRootLogin no, el acceso root remoto fue completamente bloqueado.



```
(kali㉿kali)-[~/lab1_hardening_profundo]
$ docker exec -it kali_lab1 bash
(kali㉿kali)-[~/lab1_hardening_profundo]
# ssh root@ubuntu_vulnerable_lab1
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
root@ubuntu_vulnerable_lab1's password:
Permission denied, please try again.
root@ubuntu_vulnerable_lab1's password: 
```

The image shows a Kali Linux terminal window with a dark theme. The terminal displays the command `docker exec -it kali_lab1 bash` being run. The prompt changes to `(kali㉿kali)-[~/lab1_hardening_profundo]`. Then, the command `# ssh root@ubuntu_vulnerable_lab1` is entered. The terminal shows a warning about the SSH connection not using a post-quantum key exchange algorithm. It then prompts for the password of `root@ubuntu_vulnerable_lab1`. The user enters a password, but the response is `Permission denied, please try again.`, indicating that remote root access has been blocked. The terminal window has a title bar with icons for file manager, terminal, and other applications. The background of the terminal window features a faint, stylized logo of a dragon or similar creature, and the word "KALI" is visible in the bottom right corner.

DESHABILITAR FTP



```
GNU nano 4.8 /etc/vsftpd.conf
#ls_recurse_enable=YES
#
# Customization
#
# Some of vsftpd's settings don't fit the filesystem layout by
# default.
#
# This option should be the name of a directory which is empty. Also, the
# directory should not be writable by the ftp user. This directory is used
# as a secure chroot() jail at times vsftpd does not require filesystem
# access.
secure_chroot_dir=/var/run/vsftpd/empty
#
# This string is the name of the PAM service vsftpd will use.
pam_service_name=vsftpd
#
# This option specifies the location of the RSA certificate to use for SSL
# encrypted connections.
rsa_cert_file=/etc/ssl/certs/ssl-cert-snakeoil.pem
rsa_private_key_file=/etc/ssl/private/ssl-cert-snakeoil.key
ssl_enable=NO

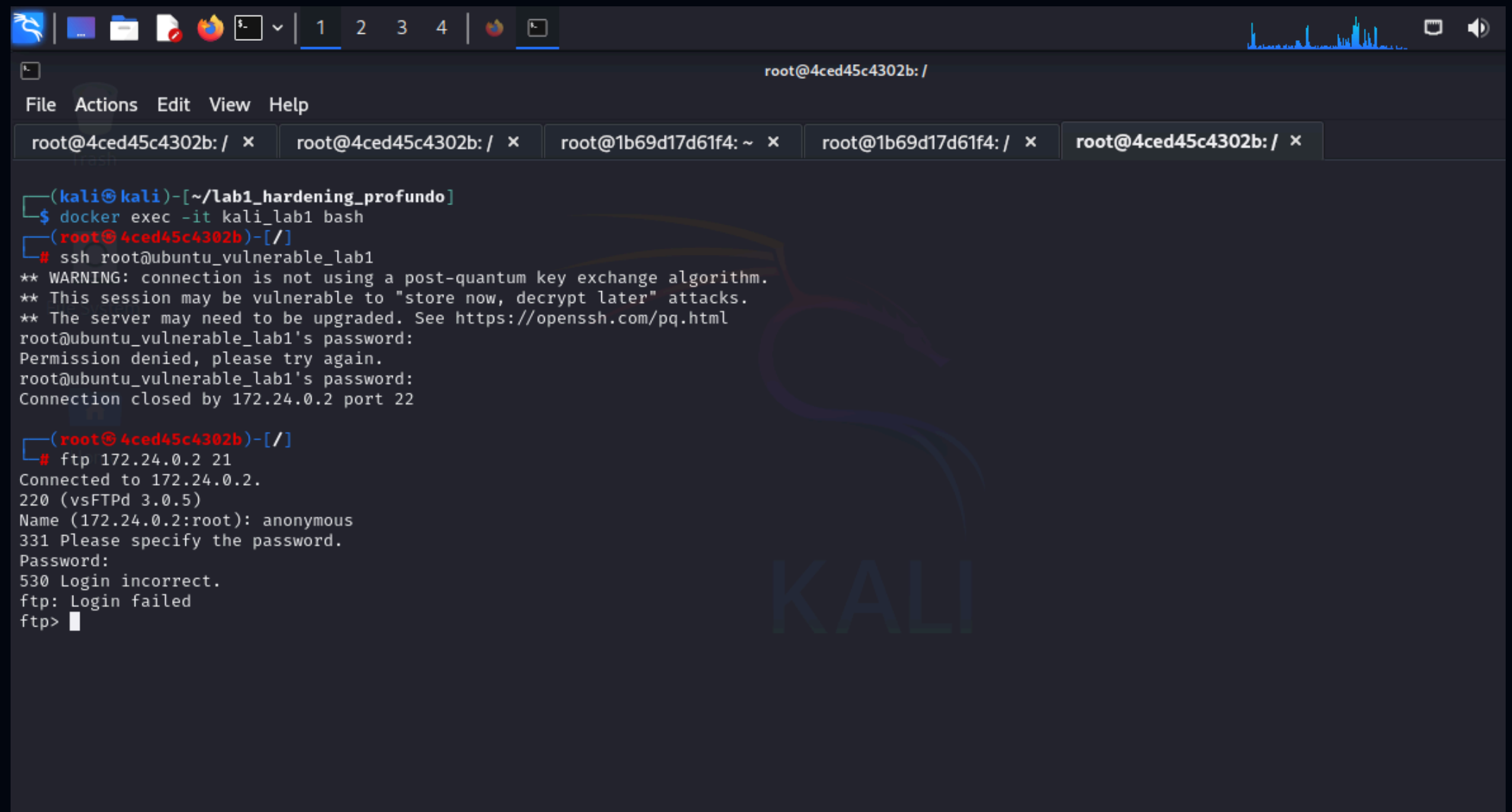
#
# Uncomment this to indicate that vsftpd use a utf8 filesystem.
#utf8_filesystem=YES
anonymous_enable=NO
```



Debido a que FTP transmite credenciales sin cifrado, se decidió deshabilitar completamente el servicio para reducir la superficie de ataque.

LOGIN FTP DENEGRADO

Tras deshabilitar el servicio FTP, los intentos de conexión desde Kali Linux fueron rechazados correctamente.

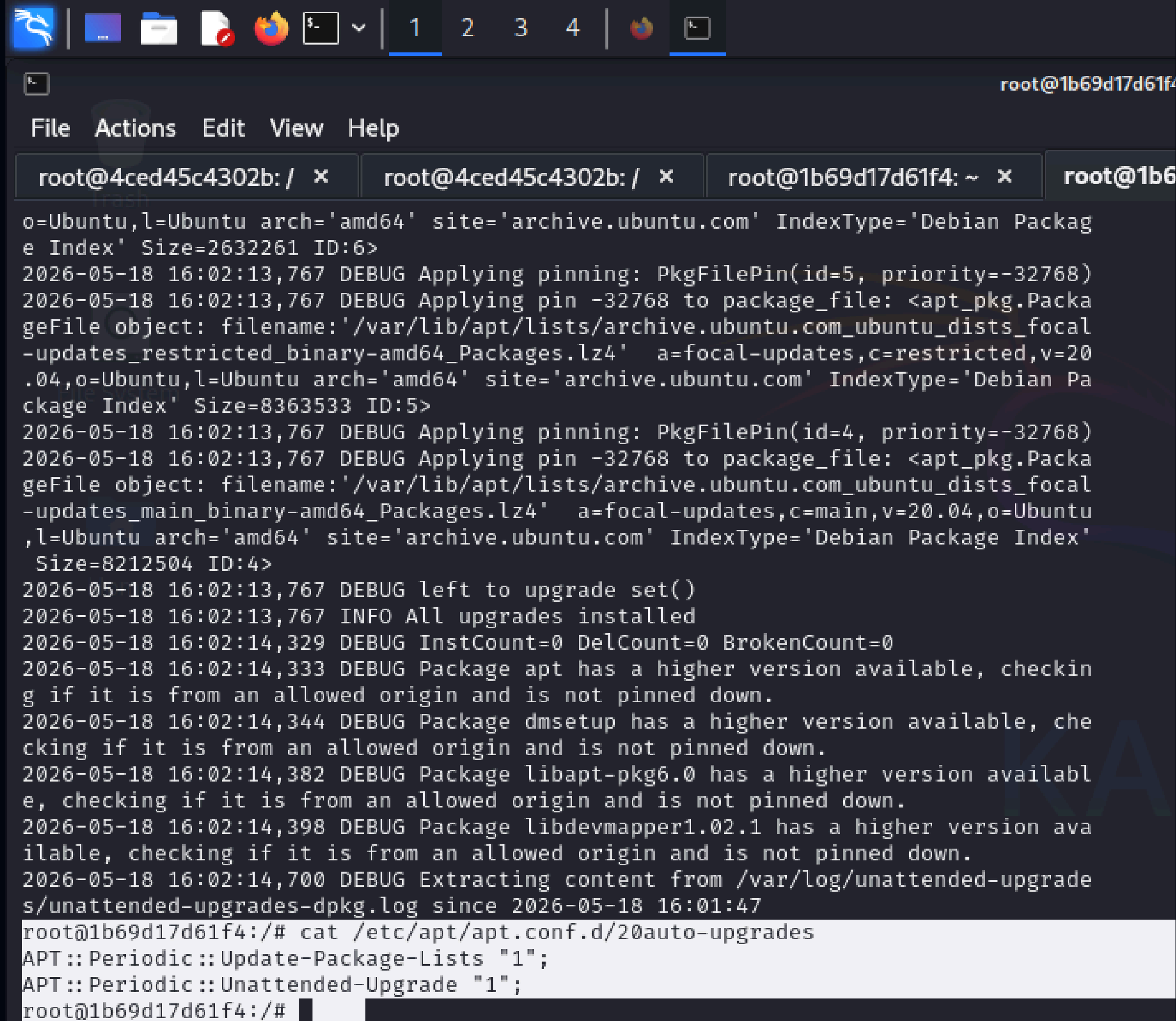


```
(kali@kali)-[~/lab1_hardening_profundo]
$ docker exec -it kali_lab1 bash
(root@4ced45c4302b)-[/]
# ssh root@ubuntu_vulnerable_lab1
** WARNING: connection is not using a post-quantum key exchange algorithm.
** This session may be vulnerable to "store now, decrypt later" attacks.
** The server may need to be upgraded. See https://openssh.com/pq.html
root@ubuntu_vulnerable_lab1's password:
Permission denied, please try again.
root@ubuntu_vulnerable_lab1's password:
Connection closed by 172.24.0.2 port 22

(root@4ced45c4302b)-[/]
# ftp 172.24.0.2 21
Connected to 172.24.0.2.
220 (vsFTPD 3.0.5)
Name (172.24.0.2:root): anonymous
331 Please specify the password.
Password:
530 Login incorrect.
ftp: Login failed
ftp>
```

AUTOMATIZACIÓN DE SEGURIDAD DEFENSIVA

- Herramienta: unattended-upgrades
- Corrección automática de vulnerabilidades
 - Sistema actualizado permanentemente
 - Reducción de riesgos de explotación



The image shows a terminal window with a dark background and a light-colored text. The window has a title bar with several icons and tabs. The main content is a log of system updates performed by the unattended-upgrades tool. The log shows the installation of updates for the 'apt' and 'dmsetup' packages, and the installation of the 'libapt-pkg6.0' and 'libdevmapper1.02.1' packages. The log also shows the extraction of content from the '/var/log/unattended-upgrades/unattended-upgrades-dpkg.log' file. The terminal prompt is 'root@1b69d17d61f4:/#'. The window title is 'root@1b69d17d61f4: ~'. The window has a menu bar with 'File', 'Actions', 'Edit', 'View', and 'Help'. The window has a tab bar with three tabs: 'root@4ced45c4302b:/', 'root@4ced45c4302b:/', and 'root@1b69d17d61f4: ~'. The window has a status bar with the text 'root@1b69d17d61f4: ~'.

```
o=Ubuntu,l=Ubuntu arch='amd64' site='archive.ubuntu.com' IndexType='Debian Packag
e Index' Size=2632261 ID:6>
2026-05-18 16:02:13,767 DEBUG Applying pinning: PkgFilePin(id=5, priority=-32768)
2026-05-18 16:02:13,767 DEBUG Applying pin -32768 to package_file: <apt_pkg.Packa
geFile object: filename:'/var/lib/apt/lists/archive.ubuntu.com_ubuntu_dists_focal
-updates_restricted_binary-amd64_Packages.lz4' a=focal-updates,c=restricted,v=20
.04,o=Ubuntu,l=Ubuntu arch='amd64' site='archive.ubuntu.com' IndexType='Debian Pa
ckage Index' Size=8363533 ID:5>
2026-05-18 16:02:13,767 DEBUG Applying pinning: PkgFilePin(id=4, priority=-32768)
2026-05-18 16:02:13,767 DEBUG Applying pin -32768 to package_file: <apt_pkg.Packa
geFile object: filename:'/var/lib/apt/lists/archive.ubuntu.com_ubuntu_dists_focal
-updates_main_binary-amd64_Packages.lz4' a=focal-updates,c=main,v=20.04,o=Ubuntu
,l=Ubuntu arch='amd64' site='archive.ubuntu.com' IndexType='Debian Package Index'
Size=8212504 ID:4>
2026-05-18 16:02:13,767 DEBUG left to upgrade set()
2026-05-18 16:02:13,767 INFO All upgrades installed
2026-05-18 16:02:14,329 DEBUG InstCount=0 DelCount=0 BrokenCount=0
2026-05-18 16:02:14,333 DEBUG Package apt has a higher version available, checkin
g if it is from an allowed origin and is not pinned down.
2026-05-18 16:02:14,344 DEBUG Package dmsetup has a higher version available, che
cking if it is from an allowed origin and is not pinned down.
2026-05-18 16:02:14,382 DEBUG Package libapt-pkg6.0 has a higher version availabl
e, checking if it is from an allowed origin and is not pinned down.
2026-05-18 16:02:14,398 DEBUG Package libdevmapper1.02.1 has a higher version ava
ilable, checking if it is from an allowed origin and is not pinned down.
2026-05-18 16:02:14,700 DEBUG Extracting content from /var/log/unattended-upgrade
s/unattended-upgrades-dpkg.log since 2026-05-18 16:01:47
root@1b69d17d61f4:/# cat /etc/apt/apt.conf.d/20auto-upgrades
APT::Periodic::Update-Package-Lists "1";
APT::Periodic::Unattended-Upgrade "1";
root@1b69d17d61f4:/#
```


ANÁLISIS DE LOGS

```
2026-05-18 16:02:13,767 DEBUG Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/archive.ubuntu.com_ubuntu_dists_focal-updates_universe_binary-amd64_Packages.lz4' a=focal-updates,c=universe,v=20.04,o=Ubuntu,l=Ubuntu arch='amd64' site='archive.ubuntu.com' IndexType='Debian Package Index' Size=2632261 ID:6>
2026-05-18 16:02:13,767 DEBUG Applying pinning: PkgFilePin(id=5, priority=-32768)
2026-05-18 16:02:13,767 DEBUG Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/archive.ubuntu.com_ubuntu_dists_focal-updates_restricted_binary-amd64_Packages.lz4' a=focal-updates,c=restricted,v=20.04,o=Ubuntu,l=Ubuntu arch='amd64' site='archive.ubuntu.com' IndexType='Debian Package Index' Size=8363533 ID:5>
2026-05-18 16:02:13,767 DEBUG Applying pinning: PkgFilePin(id=4, priority=-32768)
2026-05-18 16:02:13,767 DEBUG Applying pin -32768 to package_file: <apt_pkg.PackageFile object: filename='/var/lib/apt/lists/archive.ubuntu.com_ubuntu_dists_focal-updates_main_binary-amd64_Packages.lz4' a=focal-updates,c=main,v=20.04,o=Ubuntu,l=Ubuntu arch='amd64' site='archive.ubuntu.com' IndexType='Debian Package Index' Size=8212504 ID:4>
2026-05-18 16:02:13,767 DEBUG left to upgrade set()
2026-05-18 16:02:13,767 INFO All upgrades installed
2026-05-18 16:02:14,329 DEBUG InstCount=0 DelCount=0 BrokenCount=0
2026-05-18 16:02:14,333 DEBUG Package apt has a higher version available, checking if it is from an allowed origin and is not pinned down.
2026-05-18 16:02:14,344 DEBUG Package dmsetup has a higher version available, checking if it is from an allowed origin and is not pinned down.
2026-05-18 16:02:14,382 DEBUG Package libapt-pkg6.0 has a higher version available, checking if it is from an allowed origin and is not pinned down.
2026-05-18 16:02:14,398 DEBUG Package libdevmapper1.02.1 has a higher version available, checking if it is from an allowed origin and is not pinned down.
2026-05-18 16:02:14,700 DEBUG Extracting content from /var/log/unattended-upgrades/unattended-upgrades-dpkg.log since 2026-05-18 16:01:47
root@1b69d17d61f4:/#
```



Auditoría

Monitoreo continuo del sistema y servicios.



Capacidades Forenses

Análisis básico forense para detección de incidentes.



Se analizaron logs de autenticación y actividad del sistema para detectar eventos sospechosos y verificar cambios de seguridad.